Tsz Fei Wang

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Education

University of Waterloo | Honours Computer Science (Co-op) — Sept 2026 – May 2031 (expected)

Skills & Technologies

Languages & Frameworks: Python, Java, C, C++, C#, TypeScript, JavaScript, HTML/CSS, SQL, R, GLSL, QML, GDScript, React, Next.js, Three.js, FastAPI, PyTorch, NumPy, Pandas
Tools: Git, Linux, Bash, Docker, GitHub Actions, Firebase, Elasticsearch, CosmosDB, Vercel, Vultr, Railway, Caddy, Playwright, Postman, Jupyter, QuantConnect LEAN, Godot, Qt, KiCad, Arduino, Claude Code

Experience

Century Frontier Asset Management — Jul 2026 – Aug 2026
Quantitative Research Intern — *Shenzhen, CN*
- Designed and implemented a **point-in-time backtesting pipeline** (**Python**, **pandas**, yfinance), eliminating lookahead bias in **LLM-driven** equity research.
- Built a **model-agnostic evaluation framework** benchmarking stock-preference and inter-stock-relationship predictions across **9 LLM architectures** and **104 tickers**, isolating true signal quality from raw model capability.

Krembil Research Institute, World Community Grid — Mar 2026 – Jun 2026
Software Engineer Intern — *Toronto, ON*
- Built a **full-stack web data access portal** for Catrin (Catalogue of Transcriptional Regulatory Interactions), unifying ~**4 million** TF-TG associations across **33,000 genes** from **9** independent data sources into a single queryable interface.

Mackenzie Cybersecurity Team — Jun 2025 – Jun 2026
Team Captain, President — *Toronto, ON*
- Designed and delivered **cybersecurity** curriculum and class presentations reaching **60+ students** to raise digital-safety awareness school-wide.
- **Captained team to a national #1 finish** in CyberTitan, Canada's largest **cyberdefense** competition, coordinating strategy and technical training across the roster.

Non-Trivial Research Fellowship — Mar 2025 – Aug 2025
Machine Learning Research Fellow — *Remote*
- Built an **audio deepfake detection** model in **Python** on **Kaggle**, training and evaluating a classifier that distinguishes real from AI-synthesized speech.

Off Script – Waterloo Tech Week | *Founder & Host, 90+ registrations* — Sept 2026

Awards & Honors

Gold & Silver Medalist, IT Systems Networking – Skills Ontario (TDSB Regional & Provincial Finals) — Mar – May 2026
National Champion x2 – 1st in Canada, Cybersecurity – $1,000 scholarship — May 2025, Mar 2026
National Finalist – 2nd in Canada, Cybersecurity – $1,600 scholarship — Mar 2025
1st Place, Canada, Mirai Project – Technology Innovation in Healthcare – $400 award — Jan 2025

Projects

Physical Studio | *Python, React, Three.js, Docker* — *Won Best Use of Vultr, Hack the Valley 11 – Oct 2026*
- Web app that turns a prompt into an IKEA-style build guide for a hardware project, with a 3D model of every step and a photo check of each step on the real bench.
- Built an **MCP server** exposing CAD tools on an OpenCascade B-rep kernel for part placement, wiring and collision/clearance checks, streaming GLB scenes to a React Three Fiber viewer on Vultr.

Doppel | *Python, FastAPI, Elasticsearch* — *Hack the North – Sept 2026*
- Real-time **phishing detection** pipeline scoring newly issued TLS certificates off the public Certificate Transparency stream; built a **hybrid retrieval** system (BM25 + dense embeddings, reranked) for cross-language evidence matching, feeding an Elastic Agent Builder verdict engine gated behind human approval.

Schedule Share | *Next.js, Firebase, Gemini API* — *Sept 2026 – Present*
- Full-stack schedule-sharing platform with **20+ active users**; built a real-time **Firebase Auth/Firestore** backend for mutual schedule sharing and integrated the Gemini API for screenshot-to-calendar parsing; deployed on Vercel with automated CI/CD and cron-based data lifecycle management.

Bing Bang Dream Adventure | *Godot, GDScript* — *2nd Place, Horizons Polaris – Aug 2026*
- Rhythm game synced to **custom IMU-based flick controllers** (6-direction input); implemented the core game engine, beatmap system, and **real-time hit-timing logic** to sync note collisions with physical input.